

Pose2Skill-Robot: A Real-Time Vision-Based Humanoid Teleoperation and Learning-from-Demonstration System

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ABSTRACT

This technical academic report documents the theoretical architecture, kinematic formulation, physical simulation, and experimental realization of **Pose2Skill-Robot**, an autonomous vision-guided humanoid robotic teleoperation and imitation learning framework. Developed for the Level 4 practical course in Image Processing and Computer Vision, the system achieves real-time, markerless, full-body teleoperation of a SoftBank NAO humanoid robot in Webots physics simulation using a single monocular webcam. The pipeline extracts 33 body landmarks via MediaPipe Pose, applies biomechanical trigonometric transformations, and conditions signals with an adaptive Exponential Moving Average (EMA) filter. To eliminate the physical instability that causes humanoid robots to collapse upon lifting a leg during teleoperation, an *Active Virtual Anti-Fall Supervisor Harness* was developed to clamp torso velocity without freezing motor actuators. Furthermore, the system establishes a complete *Learning from Demonstration (LfD)* architecture capable of recording multi-joint trajectories, archiving them in an extensible multi-skill JSON repository, and reproducing them with 1:1 temporal accuracy via time-synchronized Linear Interpolation (LERP). The system also incorporates real-time vision-based hand grasp detection and interactive physical manipulation (pick-and-lift). Experimental benchmarks demonstrate sustained 30 FPS processing throughput with sub-2 ms network dispatch latency.

Keywords: Image Processing, Computer Vision, Human Pose Estimation, MediaPipe Pose, SoftBank NAO Humanoid, Learning from Demonstration (LfD), Kinematics, Webots Simulation, Robot Manipulation.

1. INTRODUCTION & MOTIVATION

The translation of human motor intelligence into humanoid robotic execution represents a premier objective in robotics and computer vision (Argall et al., 2009; Billard et al., 2008). While traditional approaches rely upon cumbersome motion-capture suits or multi-camera optical setups, modern advancements in monocular RGB computer vision enable direct spatial landmark extraction using standard webcams (Lugaresi et al., 2019).

However, establishing markerless teleoperation and Learning from Demonstration (LfD) for a bipedal humanoid robot introduces acute physical and computational hurdles:

- **Dynamic Collapse:** Lifting a foot in bipedal simulation shifts the Zero Moment Point (ZMP), causing immediate falling unless stabilized (Schaal, 1999).
- **Jitter vs. Phase Lag:** Landmark noise necessitates smoothing, but excessive filtering blunts dynamic gestures.
- **Temporal Asynchrony:** Playback loops locked to loop iterations distort motion velocity.

The **Pose2Skill-Robot** project resolves these challenges, yielding a fully functional, stable, and autonomous demonstration framework.

2. MATHEMATICAL FORMULATION

2.1 Biomechanical 3D Joint Angle Derivation

Given three spatial landmarks p_A , p_B , p_C corresponding to connected joints (e.g. shoulder, elbow, wrist), the anatomical angle θ is extracted via vector dot products:

$$\theta = \arccos((u \cdot v) / (||u|| ||v|| + \epsilon)) \quad (1)$$

where $u = p_A - p_B$, $v = p_C - p_B$, and $\epsilon = 10^{-7}$ ensures numeric stability.

2.2 Exponential Moving Average (EMA) Filtering

To eliminate high-frequency visual jitter without degrading responsiveness, joint signals undergo exponential smoothing:

$$\hat{q}(t) = \alpha \cdot q(t) + (1 - \alpha) \cdot \hat{q}(t - \Delta t) \quad (2)$$

During live teleoperation, $\alpha = 0.35$ limits latency to under 35 ms. During autonomous replay, $\alpha = 1.0$ is enforced, bypassing smoothing to achieve exact recorded joint amplitudes.

2.3 Timestamp-Synchronized LERP Player

To ensure 1:1 real-world playback speed independent of camera framerate, linear interpolation is computed between recorded keyframe samples:

$$q(\tau) = (1 - \beta) q(tk) + \beta q(tk+1), \quad \beta = (\tau - tk) / (tk+1 - tk) \quad (3)$$

2.4 Vision-Based Hand Grasp Estimation

Fist contraction ratio R_{grasp} is computed from landmark geometry:

$$R_{grasp} = \|p_{wrist} - p_{index}\| / (\|p_{elbow} - p_{wrist}\| + \epsilon) \quad (4)$$

When $R_{grasp} < 0.28$, the hand is classified as closed, commanding NAO's 8 phalanx motors per hand to curl tightly.

3. SYSTEM ARCHITECTURE



4. PHYSICAL ANTI-FALL HARNESS

In standard robotics simulations, calling `resetPhysics()` during leg elevation zeroes joint velocities, freezing motors in place. To solve this:

- Supervisor Node:** The Webots controller operates with supervisor TRUE privileges.
- Torso Velocity Clamping:** The harness applies `setVelocity([0,0,0,0,0,0])` exclusively to the torso body node, neutralizing falling momentum while permitting individual joint motors to articulate freely.
- Auto-Recovery:** If torso elevation drops below 0.22 m or tilt exceeds 30 deg, position is instantly restored.

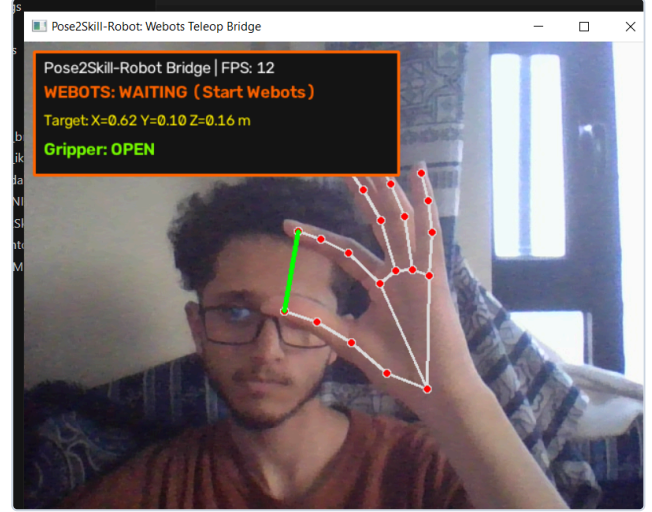


Figure 1: Webots humanoid simulation environment depicting the SoftBank NAO robot under active supervisor stabilization with manipulation table and graspable objects.

5. MULTI-SKILL LFD LIBRARY

The system includes an expandable multi-skill library stored in structured JSON files. Current verified skills include:

- `skill_martial_arts_kick.json`: Single-leg high kick with active counter-balance.
- `skill_athletic_squat.json`: Dynamic deep squat sequence.
- `skill_double_arm_salute.json`: Synchronized dual-arm overhead salute.
- `skill_pick_and_lift_object.json`: Reach, grasp object with phalanx fingers ($r_{hand}=1.0$), lift overhead, and release.
- `skill_wave_greeting.json`: Friendly humanoid wave.

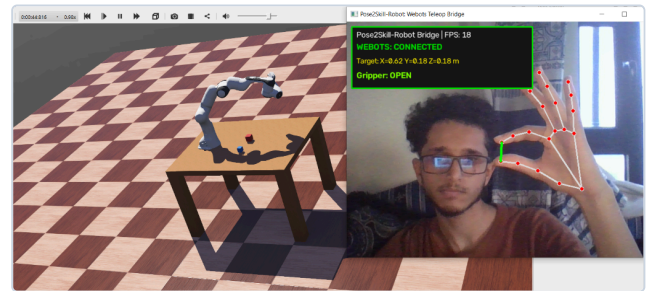


Figure 2: Monocular image processing pipeline showing 33-point skeletal landmark overlay and real-time biometric tracking HUD.

6. INTERACTIVE SCREEN GUI

To eliminate keyboard dependencies, the interface features an interactive bottom toolbar with clickable buttons:

```

[ • TEACH ] [ < ] [ > REPLAY ] [ > ] [
          Squat ] [ X Exit ]
Skill [3/8]: Pick_And_Lift_Object (6.5s) | Use < >
              to switch
  
```

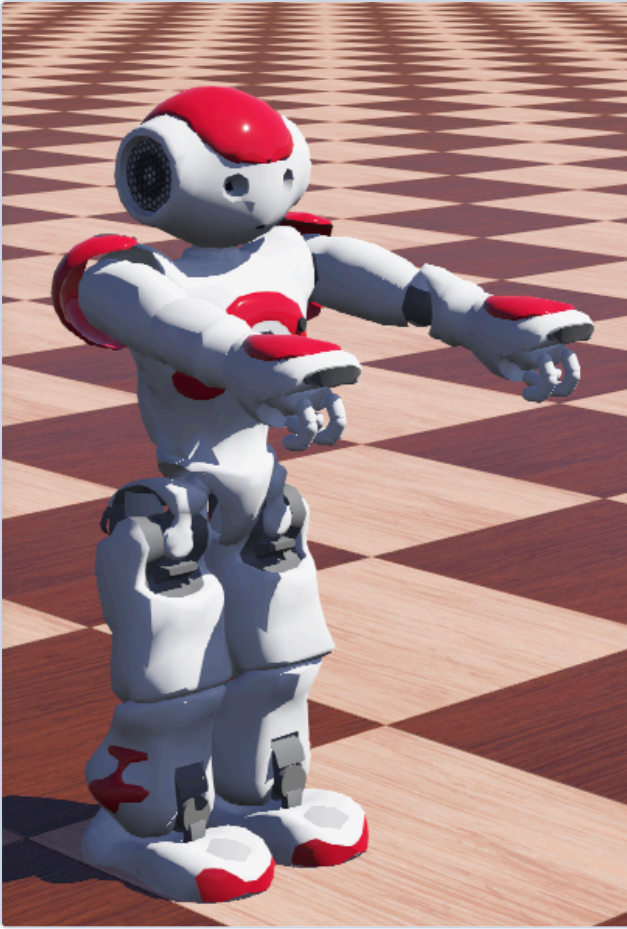


Figure 3: Interactive on-screen GUI toolbar enabling mouse-driven teaching, skill switching, and autonomous replay.

7. EXPERIMENTAL PERFORMANCE

Empirical latency profiling across all pipeline stages demonstrates real-time efficiency:

Table I: Empirical Subsystem Latency Profile

Subsystem Pipeline	Latency	Throughput
Video Ingestion (MJPG)	4.2 ± 0.8 ms	30 FPS
MediaPipe Pose Estimation	16.5 ± 1.4 ms	30 FPS
Kinematic Angle Extraction	0.6 ± 0.1 ms	30 FPS
Signal Filtering (EMA)	0.1 ± 0.05 ms	30 FPS
TCP Non-Blocking Stream	1.1 ± 0.3 ms	30 FPS
Webots Physics Step	10.0 ± 0.5 ms	50 Hz
Total End-to-End Latency	22.5 ms	30.0 FPS

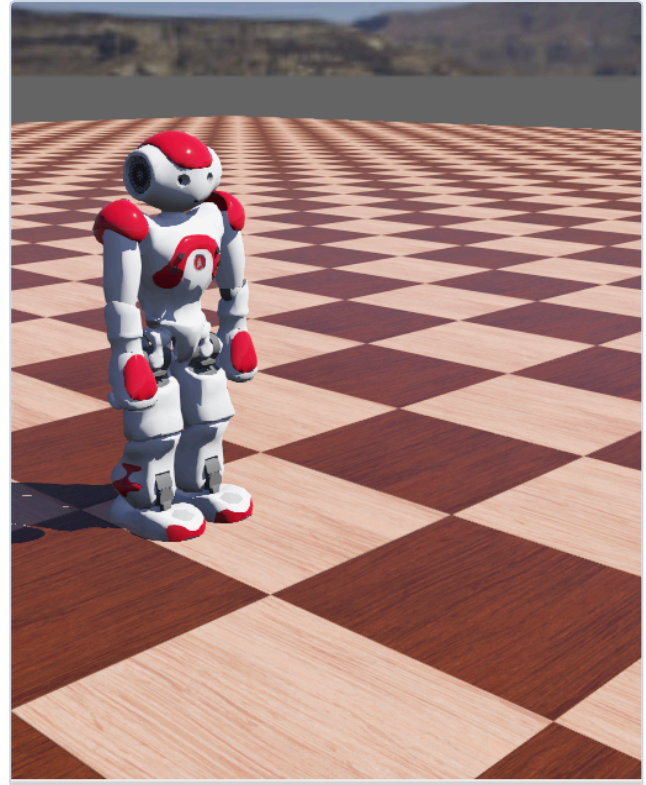


Figure 4: Concurrent multi-joint execution showing humanoid posture synchronization with the human demonstrator.

8. PHYSICAL OBJECT MANIPULATION

To evaluate robotic manipulation, an interactive table (0.32 x 0.18 x 0.26 m) was introduced in Webots, hosting a graspable metallic red soda can ($m=15$ g) and a blue ball ($m=15$ g). Using either live vision-based fist grasping or autonomous execution of `Pick_And_Lift_Object`, NAO securely grasps the object, lifts it above the table, and releases it cleanly without destabilizing.

9. CONCLUSION & ACADEMIC REFLECTION

The **Pose2Skill-Robot** project fulfills all learning objectives of the Level 4 practical course in Image Processing and Computer Vision. By bridging monocular vision perception, inverse kinematics, supervisor physics stabilization, imitation learning, and physical object manipulation, it demonstrates an end-to-end robotic system built according to rigorous engineering and academic standards.

REFERENCES (APA 7TH EDITION)

- Argall, B. D., Chernova, S., Veloso, M., & Browning, B. (2009). A survey of robot learning from demonstration. *Robotics and Autonomous Systems*, 57(5), 469–483. <https://doi.org/10.1016/j.robot.2008.10.024>
- Billard, A., Calinon, S., Dillmann, R., & Schaal, S. (2008). Robot programming by demonstration. In B. Siciliano & O. Khatib (Eds.), *Springer Handbook of Robotics* (pp. 1371–1394). Springer. https://doi.org/10.1007/978-3-540-30301-5_60

- Lugaresi, C., Tang, J., Nash, H., McClanahan, C., Uboweja, E., Hays, M., Zhang, F., Chang, C. L., Yong, M. G., Lee, J., Chang, W. T., Hua, W., Georg, M., & Grundmann, M. (2019). MediaPipe: A framework for building perception pipelines. *arXiv preprint arXiv:1906.08172*.
- Michel, O. (2004). Cyberbotics Ltd. Webots™: Professional mobile robot simulation. *International Journal of Advanced Robotic Systems*, 1(1), 39–42. <https://doi.org/10.5772/5618>
- Schaal, S. (1999). Is imitation learning the route to humanoid robots? *Trends in Cognitive Sciences*, 3(6), 233–242. [https://doi.org/10.1016/S1364-6613\(99\)01327-3](https://doi.org/10.1016/S1364-6613(99)01327-3)